

Filter Dataset

Filter the records used across all tabs.

Age Range

1864

BMI Range

15.9653.13

Smoker Status

yesno

Regions

southwestsoutheastsouthwestsoutheastnorthwestnortheast

Showing 1,338 of 1,338 people (100.0%)

Medical Insurance Cost & Statistics Lab

Explore demographic trends, run statistical tests, and forecast individual medical costs.

1. Data Exploration (EDA)2. Hypothesis Testing3. Cost Estimator & Model Health

1. General Summary Statistics

	Feature	Average (Mean)	Typical Middle (Median)	Spread (Std Dev)	Lowest	Highest
0	age	39.21	39	14.05	18	64
1	bmi	30.66	30.4	6.1	15.96	53.13
2	children	1.09	1	1.21	0	5
3	charges	13270.42	9382.03	12110.01	1121.87	63770.43

Summary Takeaway: The average medical charge for this subset is 13,270.42 **, while the typical middle individual (median) pays **9,382.03. A large gap between mean and median indicates that high medical outliers pull up the overall average.

2. Visual Distribution & Relationships

Feature to Inspect

charges

Compare with Charges (X-axis)

age

Split / Color by

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Distribution of Charges (Grouped by smoker)

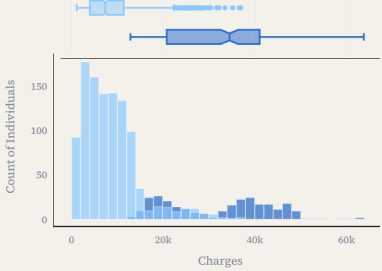


Chart Insight: Shows the spread of charges. The horizontal box on top indicates the range and median for each group.

Age vs. Charges

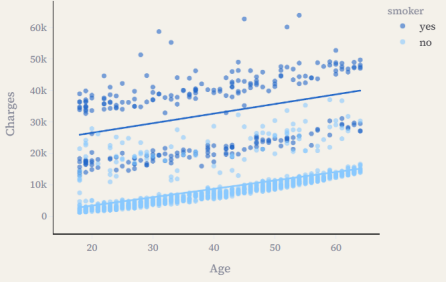


Chart Insight: Upward-sloping trendlines suggest an increase in age directly associates with higher medical bills.

